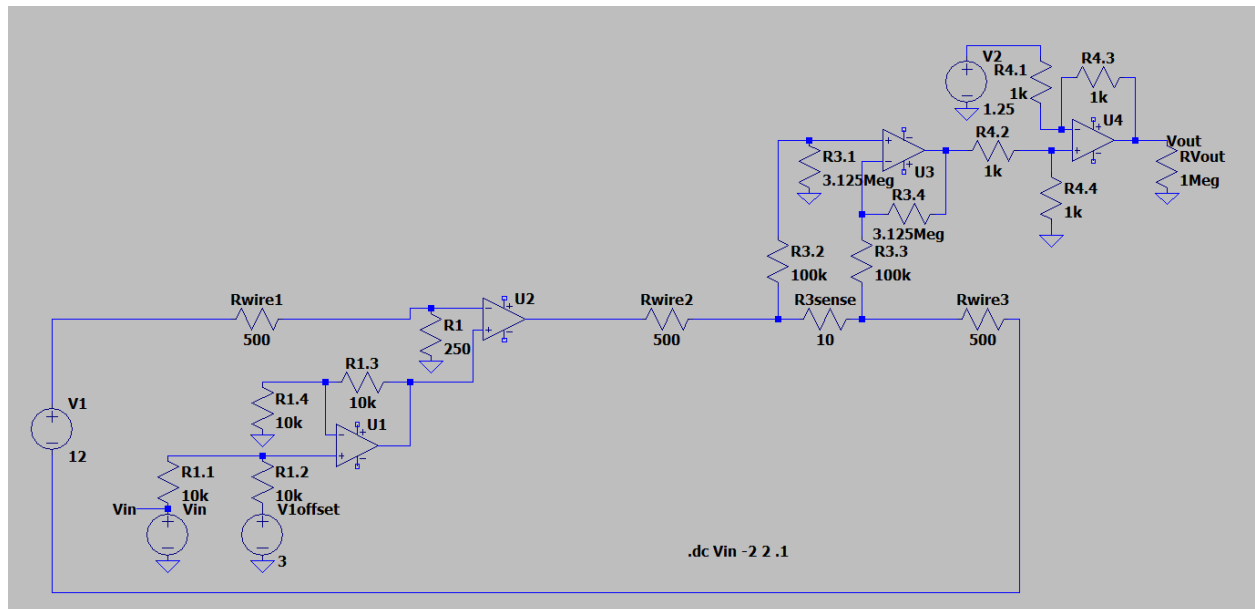
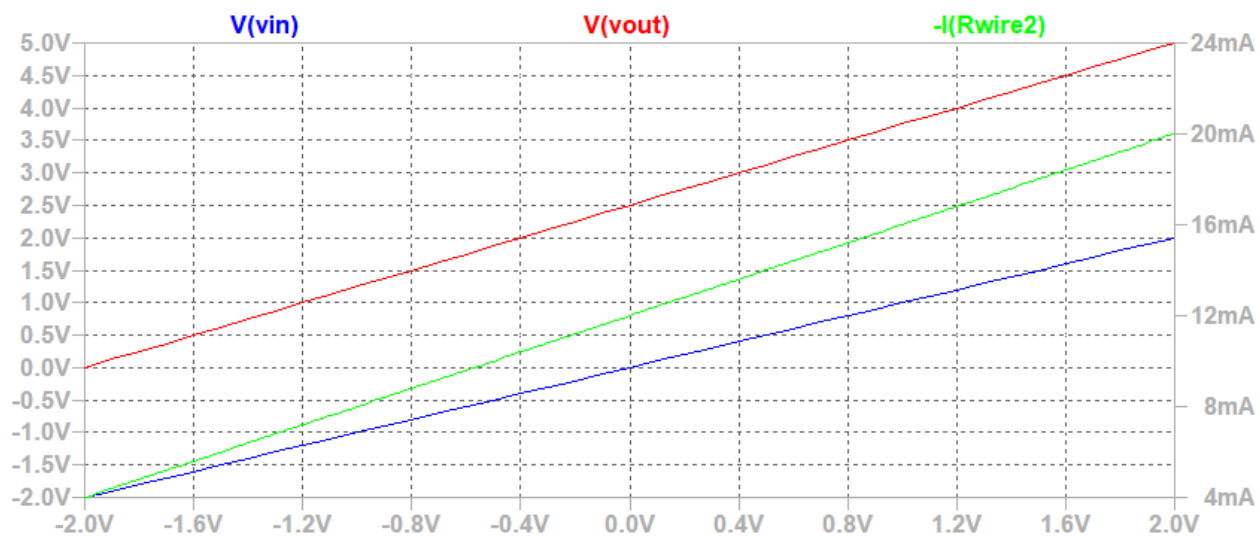






P2) Book Problem 2.21, (-4050: part a only, -6050: both a and b). The target solution for i_o in the book is wrong, it should be: $i_o = A_2 v_2 - A_1 v_1 - v_L / R_o$ instead. Find the condition where $R_o \rightarrow \infty$ which also causes $A_2 = A_1 = A$. Assume ideal op amps.

(a) Given that the circuit of Fig. P2.21 yields $i_o = A(v_2 - v_1) - (1/R_o)v_L$, find expressions for A and R_o , as well as the condition among the resistances that yields $R_o = \infty$.

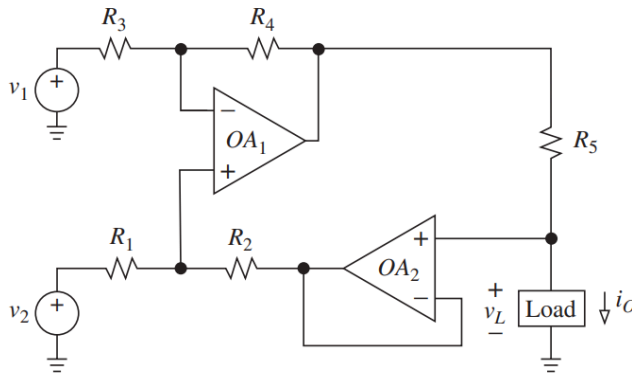


FIGURE P2.21

$$\begin{aligned}
 V_n &= (V_L - V_2) R_1 / (R_1 + R_2) + V_2 \\
 V_n &= (V_{op} - V_1) R_3 / (R_3 + R_4) + V_1 \\
 (V_L - V_2) R_1 / (R_1 + R_2) + V_2 &= (V_{op} - V_1) R_3 / (R_3 + R_4) + V_1 \\
 V_{op} - V_L &= i_o R_5 \\
 V_{op} &= i_o R_5 + V_L \\
 (V_L - V_2) R_1 / (R_1 + R_2) + V_2 &= (i_o R_5 + V_L - V_1) R_3 / (R_3 + R_4) + V_1 \\
 V_L (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4))) + V_1 (-1 + (R_3 / (R_3 + R_4))) + V_2 (1 - R_1 / (R_1 + R_2)) &= i_o (R_3 R_5 / (R_3 + R_4)) \\
 i_o &= V_L ((R_3 + R_4) / (R_3 R_5)) (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4))) + \\
 &\quad V_1 ((R_3 + R_4) / (R_3 R_5)) (-1 + (R_3 / (R_3 + R_4))) + V_2 ((R_3 + R_4) / (R_3 R_5)) (1 - R_1 / (R_1 + R_2)) \\
 i_o &= A_2 V_2 - A_1 V_1 - V_L / R_o \\
 i_o &= \\
 &\quad V_2 ((R_3 + R_4) / (R_3 R_5)) (1 - (R_1 / (R_1 + R_2))) \\
 &\quad - V_1 ((R_3 + R_4) / (R_3 R_5)) (1 - (R_3 / (R_3 + R_4))) \\
 &\quad - V_L (- (R_3 + R_4) / (R_3 R_5)) (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4))) \\
 A_2 &= ((R_3 + R_4) / (R_3 R_5)) (1 - R_1 / (R_1 + R_2)) = ((R_3 + R_4) / (R_3 R_5)) (R_2 / (R_1 + R_2)) \\
 A_2 &= (R_2 (R_3 + R_4)) / ((R_3 R_5) (R_1 + R_2)) \\
 A_1 &= ((R_3 + R_4) / (R_3 R_5)) (1 - (R_3 / (R_3 + R_4))) = ((R_3 + R_4) / (R_3 R_5)) (R_4 / (R_3 + R_4)) \\
 A_1 &= R_4 / (R_3 R_5) \\
 R_o &= -1 / (((R_3 + R_4) / (R_3 R_5)) (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4)))) \\
 R_o &= -(R_3 R_5) / (R_3 + R_4) / (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4))) \\
 \text{for } R_o \neq \infty: \\
 A_2 &= (R_2 (R_3 + R_4)) / ((R_3 R_5) (R_1 + R_2)) \\
 A_1 &= R_4 / (R_3 R_5) \\
 R_o &= -(R_3 R_5) / (R_3 + R_4) / (R_1 / (R_1 + R_2) - (R_3 / (R_3 + R_4)))
 \end{aligned}$$

for $R_o \rightarrow \infty$: $R_2/R_1 = R_4/R_3$:

$$A_2 = (R_2(R_3+R_4))/((R_3 \cdot R_5) \cdot (R_1+R_2)) = (R_2(R_1+R_2)) / ((R_1 \cdot R_5) \cdot (R_1+R_2))$$

$$A_2 = R_2/(R_1 \cdot R_5)$$

$$A_1 = R_4/(R_3 \cdot R_5) = R_2/(R_1 \cdot R_5) = A_2$$

$$A = R_2/(R_1 \cdot R_5)$$

$$i_o = R_2/(R_1 \cdot R_5) \cdot (v_2 - v_1)$$

(b) Discuss the effect of using 1% resistances.

R_o is no longer infinite, A_1 no longer exactly equals A_2

worst case: $1.01R_2/0.99R_1 = 0.99R_4/1.01R_3 \approx R_2/R_1 = R_4/R_3$ (0.961)

$$R_o = -(R_3 \cdot R_5)/(R_3+R_4) / (R_1/(R_1+R_2) - (R_3/(R_3+R_4)))$$

$$R_o = -(R_3 \cdot R_5)/(R_3+R_4) / (R_3/(R_3+R_4) - (R_3/(R_3+R_4)))$$

$$R_o = -(R_3 \cdot R_5)/(R_3+R_4) / (R_3/(R_3+R_4) - (0.961)(R_3/(R_3+R_4)))$$

$$R_o = -(R_3 \cdot R_5)/(R_3+R_4) / (0.039)R_3/(R_3+R_4)$$

$$R_o = -(25.641)R_3 \cdot R_5/(R_3+R_4) / R_3/(R_3+R_4)$$

$$R_o = -(25.641)R_5$$

P3) Book Problem 2.25. Assume ideal op amps.

Find the gain as well as the output impedance of the current amplifier of Fig. P2.25.

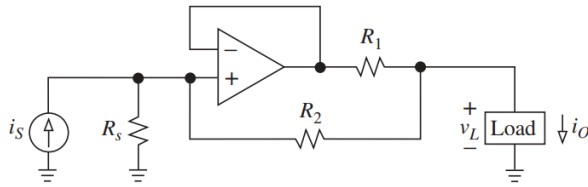


FIGURE P2.25

$$V_n = V_{R_s} = V_+ = V_- = V_{op}$$

$$i_s = i_{R_s} + i_{R_2}$$

$$i_o = i_{R_1} + i_{R_2} = (V_n - V_L)/R_1 + (V_n - V_L)/R_2 = (V_n - V_L)(R_1 + R_2)/(R_1 R_2)$$

$$i_s = i_{R_s} + i_{R_2} = V_n/R_s + (V_n - V_L)/R_2$$

$$i_s R_s R_2 + V_L R_s = V_n R_2 + V_n R_s$$

$$V_n = (i_s R_s R_2 + V_L R_s) / (R_2 + R_s) = R_2 R_s / (R_2 + R_s) (i_s + V_L / R_2)$$

$$i_o = ((R_2 R_s / (R_2 + R_s) (i_s + V_L / R_2)) - V_L) (R_1 + R_2) / (R_1 R_2)$$

$$i_o = (R_1 + R_2) / (R_1 R_2) * R_2 R_s / (R_2 + R_s) * i_s + (R_1 + R_2) / (R_1 R_2) * (R_s / (R_2 + R_s) - 1) * V_L$$

$$i_o = A i_s + V_L / R_o$$

$$A = (R_1 + R_2) / (R_1 R_2) * R_2 R_s / (R_2 + R_s)$$

$$R_o = 1 / ((R_1 + R_2) / (R_1 R_2) * (R_s / (R_2 + R_s) - 1))$$

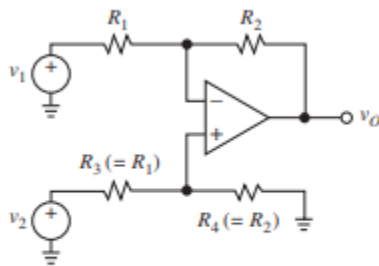
$$R_o = (R_1 R_2) / (R_1 + R_2) / (R_2 / (R_2 + R_s))$$

$$R_o = (R_1) / (R_1 + R_2) / (1 / (R_2 + R_s))$$

$$R_o = ((R_2 + R_s) R_1) / (R_1 + R_2)$$

P4) Book Problem 2.34. Assume ideal op amps.

If the actual resistance values in Fig. 2.13a are found to be $R_1 = 1.01 \text{ k}$, $R_2 = 99.7 \text{ k}$, $R_3 = 0.995 \text{ k}$, and $R_4 = 102 \text{ k}$, estimate A_{dm} , A_{cm} , and $CMRR_{dB}$.



(a)

FIGURE 2.13

$$V_o = \frac{R_4}{R_1} \frac{(R_1 + R_2)}{(R_3 + R_4)} V_2 - \frac{R_2}{R_1} V_1 = 98.749578827 V_2 - 98.71128712871 V_1$$

$$V_{DM} = V_2 - V_1 = 0.0367075398738$$

$$V_{CM} = (V_1 + V_2)/2 = 98.7312250571$$

$$CMRR_{dB} = 20 \log_{10}(100/0.0367075398738) = 68.7048944156$$

$$A_{DM} = 0.03670754 \text{ V/V}$$

$$A_{CM} = 98.73123 \text{ V/V}$$

$$CMRR_{dB} = 68.70489 \text{ dB}$$

P5) -6050 Only: Book Problem 2.39. Note that the problem defines ACM and ADM in terms of short-circuit load current, implying that your analysis can be done assuming $R_L = 0 \Omega$. If implemented with $R_1 = R_2 = R_3 = R_4$ and ideal op amps, the circuit of Fig. P2.21 gives $i_O = (v_2 - v_1)/R_5$ regardless of the load. As such, it is a difference amplifier, and therefore prone to CMRR limitations if the resistances are mismatched and op amps have $a = \infty$. For the present circuit we define $A_{dm} = i_{DM}(SC)/v_{DM}$ and $A_{cm} = i_{CM}(SC)/v_{CM}$, where $i_{DM}(SC)$ and $i_{CM}(SC)$ are, respectively, the short-circuit differential-mode and common-mode output currents, and v_{DM} and v_{CM} are as defined in Eq. (2.21).

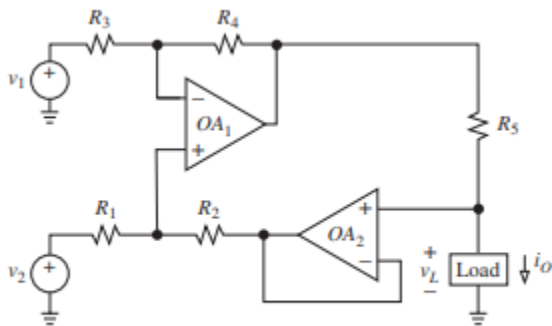


FIGURE P2.21

(a) Investigate the effect on the CMRR if we use 1% resistors and otherwise ideal op amps.

From earlier we have:

for $R_o \neq \infty$:

$$A_2 = (R_2(R_3+R_4))/((R_3 \cdot R_5)(R_1+R_2))$$

$$A_1 = R_4/(R_3 \cdot R_5)$$

$$R_o = -(R_3 \cdot R_5)/(R_3+R_4) / (R_1/(R_1+R_2) - (R_3/(R_3+R_4)))$$

for $R_o \rightarrow \infty$: $R_2/R_1 = R_4/R_3$:

$$A = R_2/(R_1 \cdot R_5)$$

$$i_o = R_2/(R_1 \cdot R_5) \cdot (v_2 - v_1)$$

$$V_L = 0$$

$$i_o = V_{op}/R_5$$

$$V_2 = 0:$$

$$V_+ = 0$$

$$V_- = (V_{op} - V_1)/(2-s) + V_1$$

$$V_- = -V_{op}/a$$

$$-V_{op}/a = (V_{op} - V_1)/(2-s) + V_1$$

$$V_{op} = -(1-s)V_1/((2-s)/a + 1)$$

$$V_1 = 0:$$

$$V_+ = V_2/2$$

$$V_- = V_{op} R/(R + R(1-s)) = V_{op}/(2-s)$$

$$V_{op} = a(V_+ - V_-) = a(V_2/2 - V_-)$$

$$V_- = V_2/2 - V_{op}/a$$

$$V_- = V_{op}/(2-s)$$

$$V_2/2 - V_{op}/a = V_{op}/(2-s)$$

$$V_{op} = (2-s)V_2/(2 + 2(2-s)/a)$$

$$V_{op} = V_2 (1-s/2)/((2-s)/a + 1)$$

total:

$$V_{op} = -(1-s)V_1/((2-s)/a + 1) + V_2 (1-s/2)/((2-s)/a + 1)$$

$$V_{op} = 1/((2-s)/a + 1) ((1-s/2) V_2 - (1-s) V_1)$$

Common mode

$$V_{op} = 1/((2-s)/a + 1) ((1-s/2) V_{CM} - (1-s) V_{CM}) = V_{CM} (s/2)/((2-s)/a + 1)$$

$$i_{CM} = V_{CM} (s/2)/((2-s)/a + 1) / R_5$$

$$A_{CM} = ((s/2)/((2-s)/a + 1)) / R_5$$

Differential Mode:

$$V_{op} = 1/((2-s)/a + 1) ((1-s/2) V_{DM}/2 - (1-s) (-V_{DM})/2)$$

$$V_{op} = 1/((2-s)/a + 1) ((1-s/2 + 1-s) V_{DM}/2) = 1/((2-s)/a + 1) ((2-3s/4) V_{DM})$$

$$V_{op} = V_{DM} (2 - 3s/4)/((2-s)/a + 1)$$

$$i_{DM} = V_{DM} (2 - 3s/4)/((2-s)/a + 1) / R_5$$

$$A_{DM} = ((2 - 3s/4)/((2-s)/a + 1)) / R_5$$

$$CMMR = 20\log(A_{DM}/A_{CM}) = 20\log(((2 - 3s/4)/((2-s)/a + 1)) / R_5) / ((s/2)/((2-s)/a + 1)) / R_5) =$$

$$CMMR = 20\log(2/s - 1.5) \approx 20\log(2/s)$$

$$\text{with } \pm 1\%, s=4\%, CMMR = 33.9794 \text{ dB}$$

(b) Investigate the effect of using perfectly matched resistors but op amps with gains of $a = 10^3$ V/V.

CMRR is about rejecting noise common to both inputs, the amplification would change how much the portion that's accepted is amplified, not how much is accepted, so CMRR = inf